

CO5420: Neural Networks and Deep Learning

Project Draft Plan

Air Pollution Forecasting Using Temporal Neural Networks

1. Group Details

Course	CO5420: Neural Networks and Deep Learning
Project Title	Air Pollution Forecasting Using Temporal Neural Networks
Batch	E23 batch
Group Members	Member 1: E/23/273 Member 2: E/23/282 Member 3: E/23/256 Member 4: E/23/011 Member 5: E/23/392

2. Motivation / Problem Definition

Air pollution is a serious public health concern, particularly in rapidly urbanizing cities. Fine particulate matter (PM_{2.5}) is one of the most harmful pollutants due to its ability to penetrate deep into the lungs, and accurate short-term forecasts of PM_{2.5} concentrations can directly support health advisory systems and environmental policy decisions.

Traditional statistical methods such as ARIMA can capture linear temporal dependencies to some extent, but they struggle with the non-linear and multivariate nature of air quality data, which is influenced simultaneously by pollutant concentrations, meteorological conditions, wind patterns, and seasonal cycles. Deep learning models, particularly those designed for sequential data, are better suited to capture these complex relationships.

This project focuses on the task of predicting the PM_{2.5} concentration at a Beijing monitoring station one hour ahead, given 24 hours of historical observations. We are drawn to this problem because it sits at the intersection of environmental science and applied deep learning, and the real-world impact of getting forecasts right is clear. It also gives us the opportunity to work with a rich, multi-feature time series dataset and explore architectures that are central to the CO 5420 course.

3. Dataset Description

The dataset used is the Beijing Multi-Site Air Quality Dataset, sourced from the UCI Machine Learning Repository. It covers hourly observations from 12 monitoring stations across Beijing, spanning March 2013 to February 2017, amounting to approximately 420,768 records in total.

Each observation includes the following features:

- Air quality measurements: PM2.5, PM10, SO2, NO2, CO, O3
- Meteorological measurements: temperature, atmospheric pressure, dew point temperature, rainfall, wind direction (categorical), and wind speed
- Temporal identifiers: year, month, day, hour

For the Kaggle task, the competition provides three files. The training set (train_raw.csv) covers the first three years of observations, while the test set (test.csv) contains flattened 24-hour input windows for which PM2.5 predictions need to be submitted. A separate file (test_raw.csv) containing the final year of raw observations is provided for the extended project work.

The dataset does contain missing values, particularly for the PM2.5 column, which will need to be handled carefully during preprocessing. Wind direction is a categorical feature and will require encoding. Feature normalization will also be necessary given the wide variation in scale across the different measurements.

4. Proposed Approach

Candidate Architectures

We plan to start with two candidate architectures:

- LSTM (Long Short-Term Memory): A natural starting point for this task given that LSTMs are designed to capture long-range dependencies in sequential data. We will use a stacked LSTM with dropout regularization as our primary baseline deep learning model.
- Temporal Convolutional Network (TCN): TCNs have shown competitive performance on sequence modelling tasks and are generally faster to train than recurrent models. We will explore this as a second architecture and compare it against the LSTM.

Depending on how these perform, we may also experiment with a hybrid CNN-LSTM or attention-based model (such as a Transformer encoder) as part of the extended work. However, we want to get solid results with simpler models first before adding complexity.

Preprocessing Plan

- Handle missing values using forward fill or linear interpolation
- Encode wind direction using sine/cosine transformation to preserve cyclical nature
- Normalize all numerical features using min-max or z-score normalization
- Construct 24-hour sliding windows from train_raw.csv to match the test format

Evaluation Metrics and Baseline

- Primary metric: Root Mean Squared Error (RMSE) - used for Kaggle ranking
- Secondary metric: Mean Absolute Error (MAE) - reported in project analysis
- Baseline: persistence model (predicting the last observed PM2.5 value as the forecast for the next hour) to give us a simple but meaningful lower bound

5. Proposed Contribution (Division of Work)

Member Name	Planned Contribution / Task	Approximate Timeline
Member 1	Data preprocessing, missing value handling, feature normalization	Week 7–8
Member 2	Training window construction, baseline persistence model implementation	Week 7–8
Member 3	LSTM model implementation, hyperparameter tuning, training on department servers	Week 8–10
Member 4	TCN model implementation and comparison with LSTM	Week 9–10
Member 5	Kaggle submission pipeline, results analysis, report writing and presentation preparation	Week 10–12

6. Generative AI Usage Statement

This project operates under Level 3 AI Collaboration, as defined in the course guidelines. During the preparation of this proposal, generative AI tools (specifically Claude by Anthropic) were used to assist with structuring ideas, refining descriptions, and reviewing the draft for clarity. All content was critically reviewed, modified, and approved by the group members. The core decisions regarding architecture selection, dataset understanding, and division of work were made by the group based on our own discussion and understanding of the course material.

Declaration: By submitting this Draft Plan, the group confirms that the content reflects the group's own understanding and decisions, and that any generative AI assistance has been disclosed above in accordance with the module's plagiarism policy.

7. References

- Zhang, S. et al. (2017). Beijing Multi-Site Air Quality Dataset. UCI Machine Learning Repository.
<https://archive.ics.uci.edu/dataset/501/beijing+multi+site+air+quality+data>
- Hochreiter, S. and Schmidhuber, J. (1997). Long Short-Term Memory. *Neural Computation*, 9(8), pp.1735–1780.
- Bai, S., Kolter, J. Z. and Koltun, V. (2018). An Empirical Evaluation of Generic Convolutional and Recurrent Networks for Sequence Modeling.
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